

# Capstone Design

Project: Verified Study Buddy

## 1. Your mission

### Your learning target

I can explain the mechanism and terminology of capstone requirements and corpus.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Human AI Workflow. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Bias and Representation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Red Team the Product. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 33: Capstone Design

Requirements, success measures, sources, risks, and tests precede building.



### CREATOR MISSION

Complete a product brief and design review.

Creator name and date:

**Write or draw your reasoning and evidence.**

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### Readiness for the connected explanation

Recall System evaluation sets. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

### This week's end goal

By the end, I can explain capstone design using a traced example and a stated limitation.

## 2. Notice the evidence

### PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use capstone design to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

**Write or draw your reasoning and evidence.**

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My prediction and the clue supporting it:

**Write or draw your reasoning and evidence.**

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## A closer explanation

A grounded-assistant capstone begins with bounded requirements and a versioned permitted corpus. A baseline can be a manual lookup process using the same evidence.

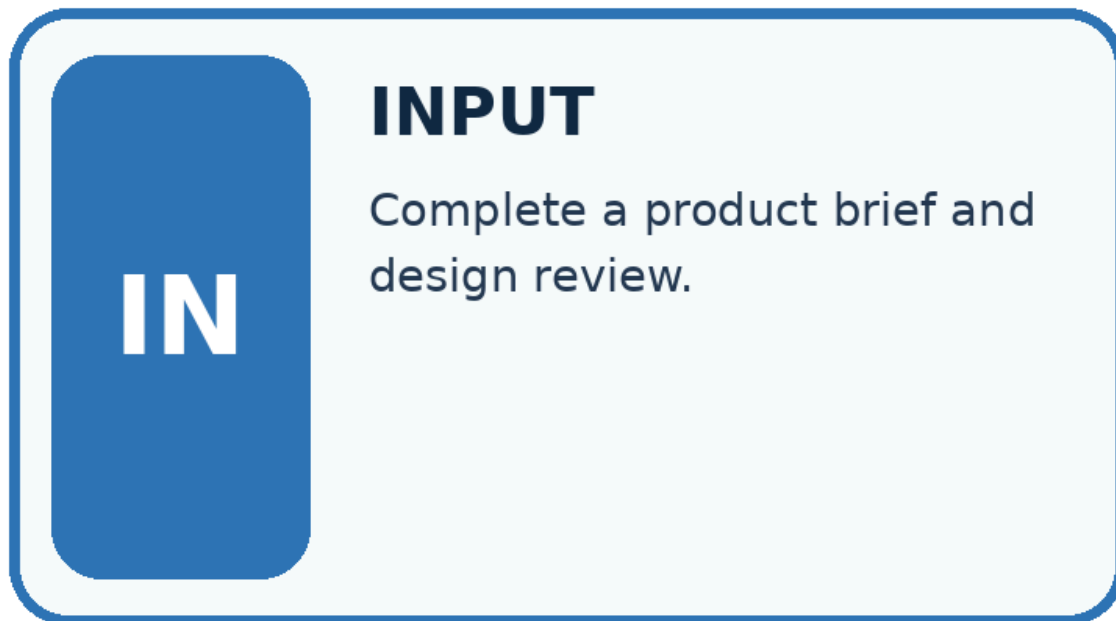
Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

## 3. See how it works

### VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



## SYSTEM

Requirements, success measures, sources, risks, and tests precede building.





## EVIDENCE

Complete a product brief and design review.

Which stage could fail even when the other two are correct? Explain:

**Write or draw your reasoning and evidence.**

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## A closer explanation

Specify the user question, source authority, access, output/fallback contract and evaluation plan. Freeze corpus IDs and split cases before implementation. Keep teacher material out of student retrieval.

## Capstone requirements and corpus — evidence trace

### Input and scope · Check or calculation · Conclusion and limit

**Input and scope:** Student room assistant corpus contains public register P, teacher-only guide T and current approval A. It must report approved room capacity or needs\_review.

**Check or calculation:** Specify the user question, source authority, access, output/fallback contract and evaluation plan. Freeze corpus IDs and split cases before implementation. Keep teacher material out of student retrieval.

**Conclusion and limit:** Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity. Reserve complete, missing and conflicting cases before development.

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The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

## Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

# 4. Compare the cases

## CASE GALLERY

### Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



**NORMAL**  
Expected learner request



**AMBIGUOUS**  
Two plausible meanings

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



## MISSING SOURCE

Evidence unavailable



## ADVERSARIAL

Rules or trust challenged

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

**Write or draw your reasoning and evidence.**

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### Change one thing in the pictured case

Change: The corpus gains a teacher-approved flower definition in V2.

Prediction: Add a flower test using V2 and record the corpus version.

Reason: The evidence boundary changed explicitly; the earlier corpus could not support this new entry.

## 5. Words that unlock the idea

### VOCABULARY LAB

#### Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



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## **SUCCESS MEASURE**

observable evidence of  
value

3

## **RISK REGISTER**

known failures and controls

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## DESIGN REVIEW

check before  
implementation

**Word · My own example · Not this / boundary**

**Word:** requirement

**My own example:** A behaviour or condition the proposed system must satisfy.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** success measure

**My own example:** An observable quantity or criterion used to decide whether a task was achieved.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** risk register

**My own example:** A record of identified risks, their consequences, mitigations and responsible reviewers.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** design review

**My own example:** An examination of a proposed system against its requirements before proceeding.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Words used in the closer explanation**

## Word · Meaning

**Word:** Corpus manifest

**Meaning:** An inventory of source IDs, versions and access roles

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**Word:** Baseline procedure

**Meaning:** A credible reference process on the same task

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## 6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

### WEEK 33 • WORKED EVIDENCE

- 1 Need: answer questions from one approved science unit.
- 2 Requirements include citations, age-fit explanation, and missing-source abstention.
- 3 Success: at least 8/10 fixed eval cases pass, with no unsupported claims.
- 4 Review architecture, privacy, provenance, and safety before building.

1. Need: answer questions from one approved science unit.
2. Requirements include citations, age-fit explanation, and missing-source abstention.
3. Success: at least 8/10 fixed eval cases pass, with no unsupported claims.
4. Review architecture, privacy, provenance, and safety before building.

Recalculate, retrace, or restate the reasoning in your own representation:

**Write or draw your reasoning and evidence.**

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CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

**A closer explanation**

Student room assistant corpus contains public register P, teacher-only guide T and current approval A. It must report approved room capacity or needs\_review.

Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity. Reserve complete, missing and conflicting cases before development.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

**Learn from the worked case**

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

**A pictorial worked case: Capstone Design**

Plan a bounded study buddy around a small approved vocabulary corpus.

**The facts we will use**

Corpus V1 defines roots, stem and leaf. It contains no plant disease treatment. Success requires a source reference; missing definitions require a clear fallback.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

## Capstone Design

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Define contract

Answer only glossary-definition requests supported by V1.



#### 2. Plan normal test

"What is a stem?" must point to V1's stem entry.



#### 3. Plan missing test

"How do I cure this disease?" must produce an out-of-scope response, not a treatment guess.

### CHANGE ONE THING

The corpus gains a teacher-approved flower definition in V2.

Explain the changed result using the matching lesson.

### Read the picture one step at a time

#### Step · What happens and why

**Step:** 1. Define contract

**What happens and why:** Answer only glossary-definition requests supported by V1.

**Step:** 2. Plan normal test

**What happens and why:** "What is a stem?" must point to V1's stem entry.

### Step: 3. Plan missing test

**What happens and why:** “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

Read the picture in order. “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

### What this example does not establish

A capstone design is a testable plan; its proposed behaviours are not verified outcomes until checked.

## 7. Find and repair a mistake

### ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



### COMPARE • DIAGNOSE

Weak idea: Building first reveals the requirements automatically. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

**Evidence record · Expected · Actual · Likely cause / next test**

**Evidence record:** normal case

**Expected:** works

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

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**Evidence record:** boundary case

**Expected:** stated rule

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

**Evidence record:** missing input

**Expected:** safe response

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

**Evidence record:** fresh transfer

**Expected:** generalizes

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

Repair one cause. Why is this repair better than changing several things?

**Write or draw your reasoning and evidence.**

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## A closer explanation

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

**Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

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## 8. Practise together

### LOGIC STUDIO

#### Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



### SPECIFY THE SYSTEM

Define task, trusted context, constraints, schema, tools, checks, fallback, and stopping rule before trusting output.

SET task, audience, constraints, output\_schema LOAD approved\_context or retrieval\_result CALL model or allowed\_tool VALIDATE fields, claims, citations, and safety IF evidence missing: ask, retrieve, or abstain RETURN useful output + source trail + limitations

Trace one input. Which line changes the state or chooses a path?

**Write or draw your reasoning and evidence.**

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### A closer explanation

A complete capstone plan is not evidence that the implementation meets it; evaluate after building.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

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### Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Capstone requirements and corpus”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

#### Plan · Do · Check

**Plan:** What is given? Which rule or procedure applies?

**Do:** Record each intermediate step.

**Check:** Does the result follow? What remains unknown?

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Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

### Check your reasoning

Use the worked case for Capstone requirements and corpus. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

### Explain the picture

What should the project plan record before the new flower test? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

**My explanation using a pictured step and the given facts**

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This is supported practice. Keep the separately named independent case for an attempt before feedback.

## 9. Run the investigation

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



### **PREDICT → RUN → VERIFY → EXPLAIN**

Write the brief, architecture, source list, schema, rubric, eval set, risks, and approval gates; peer-review for testability.

Write the brief, architecture, source list, schema, rubric, eval set, risks, and approval gates; peer-review for testability.

**Test / input · Prediction · Actual · Evidence / calculation · Change or keep?**

**Test / input:** normal

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** boundary

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** missing

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** fresh transfer

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

What did the hardest test teach you that the normal test did not?

**Write or draw your reasoning and evidence.**

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

**Plan → try → check**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

## 10. Build your understanding

## MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



### CREATOR MASTERY

1 Define • 2 Trace • 3 Build • 4 Evaluate • 5 Verify • 6 Defend limits

1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or evidence path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

**Write or draw your reasoning and evidence.**

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## 11. Connect your project

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



## VERIFIED STUDY BUDDY

Freeze the Study Buddy v1 design and acceptance tests before implementation.

THIS WEEK'S CHECKPOINT Freeze the Study Buddy v1 design and acceptance tests before implementation.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → human decision

What will you add, change, or verify in the project this week?

**Write or draw your reasoning and evidence.**

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What should remain under human control?

**Write or draw your reasoning and evidence.**

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RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent output with verified truth.

**Evidence for your project**

For your Verified Study Buddy, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

# 12. Show what you know

## INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



### INDEPENDENT CREATOR PROOF

Complete a product brief and design review.

CHALLENGE Complete a product brief and design review.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

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2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

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3. One limitation, fairness concern, or situation needing human review:

**Write or draw your reasoning and evidence.**

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SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

## 13. Study a complete case

### Week 33 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

#### The situation

Capstone: answer from one approved chapter with citations and a missing-evidence fallback.

#### The question

Define acceptance tests.

#### Worked explanation

Use answerable, ambiguous and unanswerable cases. Check correctness, support, format and fallback, plus declared permissions and privacy.

#### Explain it in your own words

Which detail supports the answer, and why does it matter?

**Write or draw your reasoning and evidence.**

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### Trace the supplied case visually

### Acceptance tests include the boundaries

#### Evidence or stage · Value or result

**Evidence or stage:** Answerable

**Value or result:** Correct cited chapter answer

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**Evidence or stage:** Ambiguous

**Value or result:** Clarifying question

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**Evidence or stage:** Unanswerable

**Value or result:** Missing-evidence fallback

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**Evidence or stage:** Permissions

**Value or result:** Respect data and action limits

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Capstone: answer from one approved chapter with citations and a missing-evidence fallback.

Define acceptance tests.

Use answerable, ambiguous and unanswerable cases. Check correctness, support, format and fallback, plus declared permissions and privacy.

## 14. Try a new case independently

### Independent case: Changed independent case

A science assistant may use only these notes: evaporation is liquid becoming vapour; condensation is vapour becoming liquid. Specify behavior for “Define evaporation”, “Explain it”, and “What is tomorrow’s weather?”.

### My independent explanation, trace and limits

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Week 33 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

The capstone goal is helps students. Rewrite it as an observable test for a passage-based study assistant.

My answer, with a trace, calculation or supporting evidence

### Write or draw your reasoning and evidence.

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### Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

### Write or draw your reasoning and evidence.

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## Connect to your project

Freeze the Study Buddy v1 design and acceptance tests before implementation.

The evidence I will keep

**Write or draw your reasoning and evidence.**

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Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

## Transfer practice from the original programme

Design a chapter assistant with one answerable, one ambiguous and one unanswerable query. State the expected behaviour and one mandatory quality criterion.

**Explain the mechanism, evidence and one limit.**

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## Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.